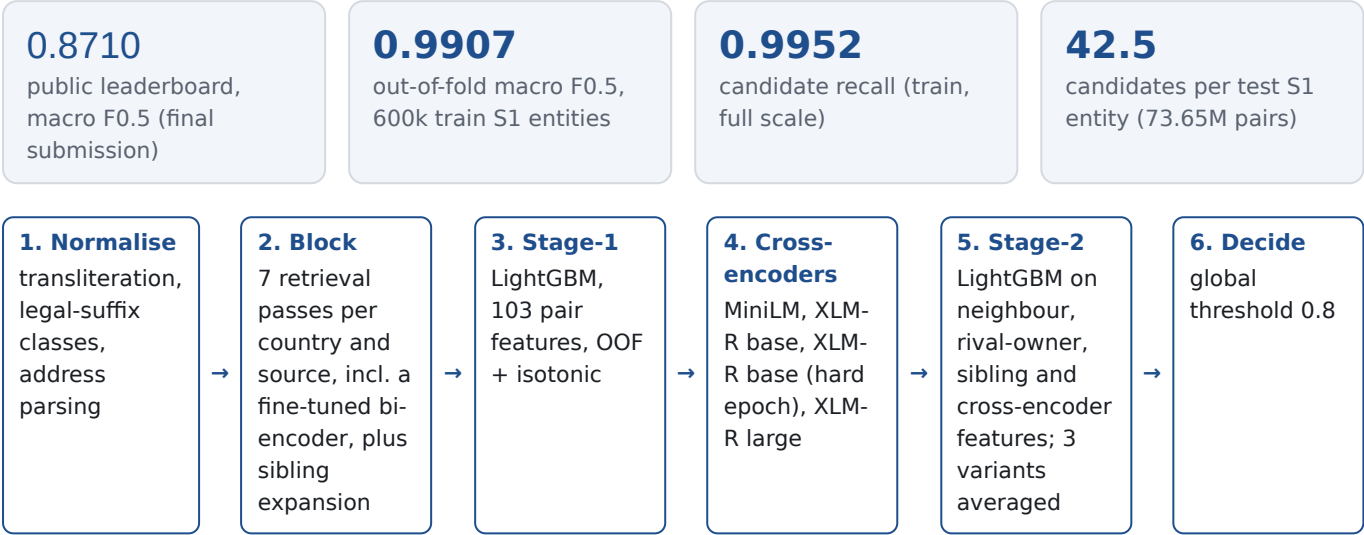


Amazon ML Challenge 2026: Business Entity Resolution

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1. Methodology

1.1 Problem analysis

For each Source-1 (S1) business, the task is to return every record in Sources 2 and 3 (S2, S3) that refers to the same business, scored by macro F0.5 per S1 entity. Our exploratory analysis of the training split (2.21M S1 entities, 10.32M S2+S3 records) shaped the design:

Finding	Value	Design consequence
S1 entities with no match	5.6% (same in US and India)	Empty predictions must be possible; a single false match on these scores 0.
Matches per S1	3.46 on average (≈ 1.7 in S2, ≈ 1.8 in S3)	Retrieve from S2 and S3 separately. An entity's matches form a cluster of near-duplicates, which we exploit.
Pool records owned by some S1	74%; 26% are distractors built from S1 names	The main source of false merges (85% of our early false-positive pairs).
Pool records matching more than one S1	0	One-owner constraint, encoded as learned rival-owner features.
Matched pairs with the same country	100%	Country used only as an equality key for blocking; never as a feature value.
Test set	1.73M S1: India 810k, US 663k, France 259k (not in train)	No country-specific rules; validation must simulate an unseen country.

Noise in matched groups includes legal-suffix swaps (Pvt Ltd / Private Limited / LLC; SARL / SAS in test), typos, word reordering, website and hashtag forms, garbled names paired with a correct address, and **native-script names** (Devanagari, Malayalam, Kannada, Bengali) for Indian

businesses. Addresses are reordered, abbreviated (St, Rd, French R., N°), truncated, missing postcodes, or carry landmarks ("Near SBI ATM").

1.2 Overall approach

A hybrid, two-round pipeline: high-recall blocking, a gradient-boosted pair classifier, four fine-tuned multilingual cross-encoders, and a second gradient-boosted model that re-scores each pair using the other pairs around it. The central ideas are:

1. **Treat matching as a cluster problem.** An entity's true matches are near-duplicates of each other, so its confident candidates are reused as queries (*sibling expansion*) and as features (*sibling similarity*).
2. **Use the one-owner constraint as evidence.** Each pool record belongs to at most one S1, so a pair's score is compared with the strongest rival S1 claiming the same record.
3. **Detect distractors by "pool twins".** Genuine records almost always have a near-duplicate in the other source; synthetic distractors rarely do. This signal alone separates distractors with AUC 0.79.
4. **Read raw multilingual text with cross-encoders.** They match native-script and Latin-script names directly, which string metrics cannot.
5. **Validate for an unseen country.** Because France appears only in test, stage-2 features and the decision threshold were chosen by training on one training country and scoring the other (leave-country-out).

1.3 Validation protocol

- **Main CV:** a fixed random sample of 600k train S1 entities (seed 42); 5-fold GroupKFold by S1 entity; out-of-fold macro F0.5 computed with the exact challenge formula over all sampled entities, including no-match entities.
- **Leakage control:** the retrieval bi-encoder and all four cross-encoders are trained only on train S1 entities *outside* the 600k stage-2 sample, so their scores on the sample are out-of-sample when stacked.
- **Leave-country-out (LCO):** stage-2 trained on India and scored on the US, and vice versa. Used to accept or reject features, hyper-parameters and thresholds for the unseen country.
- **Acceptance rule:** a change was kept only if CV improved by ≥ 0.0005 (≥ 0.0003 for cheap changes), LCO did not worsen, and per-country test statistics (matches per S1, share of empty predictions) stayed consistent with the training distribution.

1.4 Normalisation

Identical for every country (no branches on the country value):

- Unicode transliteration of all scripts to Latin, lower-casing and punctuation clean-up; the raw text is also kept for the cross-encoders.
- **Legal-suffix classes** (e.g. *Pvt Ltd*, *Private Limited*, *LLC*, *Inc*, *SARL*, *SAS*) are stripped into a separate field, leaving a *name core*; abbreviation expansion and "doing business as" variants.
- A **consonant skeleton** and Metaphone code so that transliterated or garbled names still meet ("praaivett limittedd" $\approx$ "private limited").
- Address parsing: house and unit numbers, postcode, city guess, street words, landmarks; ordinals ("10ST"), leading zeros, NULL; French forms (N°, R. = rue, St = Saint) treated as generic abbreviations.

2. Candidate Generation / Blocking Strategy

Every S1 record is compared only with pool records carrying the same country string, and retrieval runs **separately against S2 and S3** so that both sources are always represented. Each pass returns its top-K neighbours per source; the candidate set is the union.

Pass	Representation and search	K / source
w word	TF-IDF uni- and bi-grams of name core + address core (document frequency ≤ 3000), exact sparse cosine	10
e1 embedding	paraphrase-multilingual-MiniLM-L12-v2 embedding of raw "name, address", exact GPU cosine	3
c character	character 3-gram TF-IDF of the compacted name ($df \leq 20000$)	3
a address $\rightarrow$ non-Latin	address TF-IDF against pool records whose name is in a non-Latin script (46% of early misses)	5
x1, x2 exact keys	(house number, first name token) and (Metaphone of name, city guess); key groups ≤ 30 records	all
f fine-tuned bi-encoder	MiniLM fine-tuned for retrieval on train matches (details below), exact top-K per country and source	5
s sibling expansion	word TF-IDF neighbours of the entity's confident candidates. Round 1: top 3 candidates by a cheap score; round 2: candidates with round-1 model probability ≥ 0.5 (top 5)	3

2.1 Fine-tuned retrieval bi-encoder (pass f)

- Backbone paraphrase-multilingual-MiniLM-L12-v2; input "name | address" with native scripts kept; mean pooling; 64 tokens.
- Trained on up to 6M (S1, true match) pairs from train S1 entities outside the stage-2 sample; symmetric InfoNCE with in-batch negatives (scale 20). Batches are drawn within one country to make negatives harder; other true matches of the same S1 in a batch are masked. 1 epoch, batch 1024, lr 3e-5, bf16.
- Alone, at top-10 per source, it finds 99.2% of true pairs. Added to the other passes at K = 5 it raised recall from 0.9813 to **0.9952** and the best score reachable on the candidates (oracle F0.5) from 0.9928 to 0.9986, for +4.6 candidates per S1.

2.2 How we avoided losing true matches

- **K tuned at the knee** on a 100k-S1 sample against the full pool. Plain union recall plateaus near 0.98 even at 287 candidates per S1; widening K (0.9796 recall at 65 per S1) *lowered* stage-1 F0.5 because it adds more noise than matches.
- **Targeted passes from miss analysis:** non-Latin names were reached through their address (pass a); matches far from the S1 text but close to its other matches were reached through siblings (pass s); the remaining semantic misses were reached by the learned bi-encoder (pass f).
- **Two rounds:** round 1 trains a model whose probabilities choose better siblings for round 2.

Blocking configuration (train)	Recall	Candidates / S1
word TF-IDF only, K = 5 / 10 / 40	0.924 / 0.949 / 0.970	10 / 20 / 80
direct passes (w, e1, c, a, keys)	0.9696	35.4
+ sibling expansion (cheap-score / model-chosen siblings)	0.9742 / 0.9813	37.0 / 38.5
+ fine-tuned bi-encoder pass f, K = 5 (final)	0.9952	43.3

Final candidate sets: **train 95.5M pairs** (43.3 per S1, reduction ratio ≈ 0.99999) and **test 73.65M pairs** (42.5 per S1). Every predicted match is one of these candidates, as required by `candidate_pairs.tsv`.

3. Model Architecture and Feature Engineering

3.1 Stage-1 pair classifier

LightGBM (binary, 63 leaves, learning rate 0.05, feature and bagging fraction 0.8, early stopping) on the ~ 26 M candidate pairs of the 600k-S1 sample; 5-fold GroupKFold by S1 gives out-of-fold probabilities $p1$, followed by isotonic calibration. **103 features**:

Group	Features
Name similarity	rapidfuzz ratio, token-set, token-sort, partial ratio and Jaro-Winkler on the name core; compact-name ratios; Metaphone and consonant-skeleton similarity; best score over DBA variants; acronym match; first and last token agreement.
IDF-weighted overlap	share of each side's IDF weight that is shared; IDF of the rarest shared, S1-only and candidate-only token; number of candidate-only tokens. Separates a shared <i>rare</i> word ("Roopaya") from a shared common one ("Sai", "Shree"). Largest single feature gain: +0.010 CV.
Legal suffix	state of the legal forms: both missing / equal / one missing / conflicting.
Address	token-set / sort / ratio / partial on address core and street words; house-number Jaccard, conflict and containment; primary-number state; postcode state and 3-digit prefix; city agreement; landmarks; empty-address flag.
Retrieval signals	MiniLM cosines (full text, name only); word / char / address TF-IDF cosines; rank of the pair in each blocking pass.
Context	rank and gap within the S1's candidate list; reverse rank and gap among all S1s that retrieved the same pool record; list sizes; mutual-best flag; name-core frequency (chains).
Pool twins	similarity of the pool record to its nearest neighbour in the other source. Genuine records have a twin; distractors usually do not (AUC 0.79 alone; no-match-entity F0.5 0.988 $\rightarrow$ 0.992).
Country	<code>same_country</code> only; the country value itself is never a feature.

The bi-encoder's own score and rank (`cos_f`, `r_f`) are deliberately excluded from stage 1, because stage-1 predictions on non-sample entities feed the stage-2 rival features and the bi-encoder was trained on those entities.

3.2 Cross-encoders

Four transformer cross-encoders read the two records together as `name | address` pairs (raw text, any script, 96 tokens) and output one logit. Each is fine-tuned for one epoch (AdamW, bf16) on all positives plus the 6 hardest and 2 random negatives per S1 from **200k train S1 entities disjoint from the stage-2 sample**, so their scores can be stacked without label leakage.

Model	Backbone (licence, size)	Training
CE-1	paraphrase-multilingual-MiniLM-L12-v2 (Apache-2.0, 118M)	round-1 candidates, lr 5e-5
CE-2	xlm-roberta-base (MIT, 278M)	round-2 candidates, lr 3e-5
CE-3	CE-2 + one more epoch on hard pairs	pairs the stage-1 model finds difficult, lr 1e-5
CE-4	xlm-roberta-large (MIT, 560M)	round-2 candidates, lr 1e-5

Pairs are scored when $p1 \geq 0.01$ or when found by pass f (the gate costs at most 0.0005 F0.5). Holdout pair AUC on 10k unseen S1: MiniLM 0.99972, XLM-R base 0.99986. The cross-encoders were the largest single model gain (+0.009 CV); for example a Devanagari record of "Baba Power Private Limited" and its Latin-script copy receive logit +9.0.

3.3 Stage-2 re-scoring model

A second LightGBM re-scores each candidate from the pairs around it (5-fold GroupKFold OOF, 3 seeds with raw scores averaged, then isotonic calibration). Final feature set:

- **Own evidence:** stage-1 $p1$, the cheap combined score, embedding and TF-IDF cosines, source flag (S2 / S3).
- **Entity context:** highest and second-highest $p1$ in the S1's list, gap to the best, sum of $p1$.
- **Rival owner (one-owner constraint):** $p1$ of the strongest other S1 claiming the same pool record, and the margin to it.
- **Siblings:** word and address TF-IDF similarity of the candidate to the S1's confident candidates (maximum, confidence-weighted maximum, number of siblings, number with address similarity ≥ 0.8).
- **Cross-encoder logits:** CE-1 to CE-4.

Features removed after the leave-country-out check. The bi-encoder score and rank ($\cos_f$, r_f) and the candidate-density features (list sizes, reverse-list size, counts of candidates above 0.5, rank features) raised in-country CV but transferred badly to a held-out country. Keeping them (version v10m) improved CV by +0.0025 yet *lowered* the leaderboard, with France over-matching (5.3% empty predictions against a 5.6% no-match rate). Dropping them, adding CE-4 and using threshold 0.8 gave the largest leaderboard gain of the challenge: 0.9778 $\rightarrow$ 0.9847.

3.4 Decision layer and final ensemble

- The final submission averages the stage-2 probabilities of three variants that differ only in stage-2 configuration: **v12** (keeps the density features), **v13** (LCO feature set) and **v15** (LCO feature set, `min_child_samples` 500 for stronger regularisation, the best LCO setting).
- A pair is accepted when the averaged probability is ≥ 0.8 ; an S1 with no accepted pair gets an empty prediction. The threshold was chosen on LCO (0.8–0.85 best there, while in-country CV preferred 0.70–0.75) and confirmed on the leaderboard: 0.85 scored 0.984767, 0.75 scored lower, 0.8 scored best.
- The one-owner constraint acts through the rival-owner features; an explicit post-hoc damping of non-maximal owners ($\alpha \in \{0, 0.3, 1\}$) changed LCO F0.5 by less than 0.00003, so no damping is applied.
- We also evaluated per-entity expected-F0.5 subset selection (Monte-Carlo over calibrated probabilities); it tied with or was below the global threshold, especially on no-match entities.

4. Other Relevant Information

4.1 Development history

Version	Main change	CV (OOF)	Public LB
v1-v2	normalisation, multi-pass blocking, LightGBM; + IDF-weighted name overlap	0.9597 → 0.9698	-
v3	+ non-Latin address pass, stage-2 stacking with rival-owner features	0.9727	0.957
v4	+ generic French abbreviation handling	≈0.9727	0.958
v5	+ sibling expansion and features, MiniLM cross-encoder	0.9820	0.973
v6-v7	+ model-chosen siblings, pool twins; + XLM-R base cross-encoder	0.9851	0.9768
v9	+ hard-example XLM-R epoch, 3-seed stage-2	0.9856	0.9778
v10m	+ fine-tuned bi-encoder pass (recall 0.9813 → 0.9952), all features kept	0.9881	0.9769
v13	+ XLM-R large; LCO-selected stage-2 features; threshold 0.8	0.9905	0.984738
v15	+ stage-2 <code>min_child_samples</code> 500	0.9907	0.984826
final	average of v12, v13 and v15 stage-2 probabilities, threshold 0.8	-	0.8710

Held-out-country F0.5 of the final stage-2 configuration: India 0.9914, US 0.9896 (stage-2 trained on the other country). On test, France's statistics match the training countries (≈3.2 matches per S1, 6.3% empty predictions vs a 5.6% train no-match rate).

4.2 What did not help (rejected)

- Wider blocking K: more noise than recall.
- An extra XLM-R epoch on confident French *test* pseudo-pairs: CV +0.0002 but leaderboard below v7. Rejected; **the final model uses no signal derived from test data**.
- Self-training LightGBM on unseen-country pseudo-labels: small and unstable gains.
- Per-entity sample weighting ($1/\sqrt{n}$) in stage 2; stage-2 without *p1*-derived features (LCO 0.983 / 0.979); a BGE-M3 bi-encoder (not completed).
- Stricter thresholds (0.85, 0.9) and a looser one (0.75) on the final model.

4.3 Error analysis

- **False positives:** distractors built from an S1 name ("Veis" → "Veis Incorporated"; "DENET LLC" with no address); chains and common Indian names with near-identical records under different S1s; in France, the same name at a neighbouring house number.
- **False negatives:** native-script names with truncated addresses (largely recovered by the non-Latin address pass, siblings and cross-encoders); "first token + legal form + generic word" variants when a rival S1 shares the first token; the remaining 0.5% of matches outside the candidate set.

4.4 Compliance and reproducibility

- **No external data, APIs or geocoding.** TF-IDF and IDF statistics are fitted on the unlabelled text of each split; all supervised models use training labels only.

- **Pretrained models:** paraphrase-multilingual-MiniLM-L12-v2 (Apache-2.0, 118M), xlm-roberta-base (MIT, 278M), xlm-roberta-large (MIT, 560M); LightGBM (MIT). All are $\leq 8\text{B}$ parameters and permissively licensed.
- **No country-specific logic:** the country string is used only as an equality key for blocking and as `same_country`; normalisation rules apply to all rows.
- **Reproducibility:** fixed seeds (42; stage-2 seeds 42-44). One script (`reproduce.sh`) runs every step from the raw TSVs to the submitted files: round-1 blocking and models, bi-encoder training, round-2 blocking, the three XLM-R cross-encoders, the three stage-2 refits and the final average. About 20 hours on one data-centre GPU with 32+ CPU cores and 64+ GB RAM.
- **Hardware used:** a 32-core workstation with an RTX PRO 4000; a 64-core server with 2× RTX A6000; one H200 GPU on a 192-core server.
- **Stack:** Python 3.10, pandas, numpy, scikit-learn, LightGBM 4.7, rapidfuzz, jellyfish, unidecode, PyTorch, transformers, sentence-transformers.